Aakriti Shah

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Education

MS **University of Pittsburgh**, Computational Biomedicine & Biotechnology — Aug. 2024 - **Exp. May 2027**
- **Coursework:** Genomics, Bioinformatics Programming, Comp. Bio Data and Models, Structure-Based Drug Discovery

MS **University of Southern California**, Computer Science — Aug. 2024 - May 2026
- GPA: 3.75 – *Magna Cum Laude*
- **Coursework:** Trustworthy Machine Learning, Natural Language Processing, Deep Learning and Its Applications, Algorithms, Innovation for Defense Applications, Text as Data (NLP and LLMs)

BA **Rollins College**, Computer Science & Business Management (*Honors*) — Sept. 2020 – Dec. 2023
- GPA: 3.72 – *Magna Cum Laude*
- **Coursework:** Operating Systems, Algorithms, Software Engineering, Computer Architecture, Data Science, Discrete Math, Software Engineering

Technical Skills

Programming: Python, SQL, Java, JavaScript, R, MATLAB, SLURM and HPC, Assembly (ARM), C, C++, XML, Dart, LaTeX, Swift
ML/DL Frameworks: PyTorch, TensorFlow, Hugging Face, scikit-learn, pandas, NumPy
Tools & Platforms: Git, Docker, Jupyter, Tableau, Google Cloud Platform, MongoDB, Flask, React, Node.js
Research Skills: Experiment design, data annotation, literature review, statistical analysis, scientific writing

Experience

Graduate Researcher, Barton Lab – Prof. John Barton (University of Pittsburgh) — Sept. 2026 – Ongoing
- Testing whether **protein language model (ESM-2) embeddings** of HIV-1 Env sequences improve a published antibody-neutralization predictor (Grouped Neutralization Learning) over its PCA features, on a CATNAP matrix of 806 antibodies × ∼1,150 viruses.
- Reproduced the authors' Julia model (Pearson R = 0.74) and built a paired evaluation that swaps only the virus features, with new hold-outs for unseen viruses and subtypes plus a random-projection control; extending to **novel antibody** prediction.

NSF IRES Research Fellow, Northumbria University – Newcastle, UK — May - July 2026
- Selected for the NSF-funded Citizen-Centered AI Summer Research Program, conducting human-centered AI research in an international, interdisciplinary setting, with the theme of **Identity, Privacy, Security, and Information**.
- Collaborating with researchers from the US and UK on sociotechnical AI challenges at the intersection of community, design, and society.

Group Project Manager and Developer, USC × Defense Health Agency — Jan. – May 2025
- Collaborated with Dr. David Brody of Walter Reed National Military Medical Center to design and build a three-component system, an AI triage app, drone resupply optimizer, and portable eddy current hemorrhage detector, aimed at reducing preventable TBI deaths in contested battlefield environments.
- Acted as project manager on this team of students.

Graduate Researcher, ARKAI Lab - Dr. Thai Le (Computer Science, Indiana University) — May 2025 – Ongoing
Project Paper
- Investigating targeted unlearning in LLMs as a **privacy-preserving** mechanism, analyzing how selectively removing sensitive information mitigates risks of unauthorized data

exposure.

- Examining knowledge entanglement in unlearned LLMs to assess information leakage vulnerabilities, evaluating resistance to **adversarial extraction attacks** and membership inference.

Graduate Researcher, AIMed Lab - Dr. Rakesh Shiradkar (Computer Science, Indiana University) May 2025 – Ongoing
Project

- Conducting computational pathology research on prostate **cancer detection** through automated tissue compartment and Gleason pattern segmentation in H&E-stained whole slide images.
- Extracted patches from and annotated glands from H&E-stained whole slide images using *QuPath*.
- Developed a novel postprocessing pipeline, incorporating lumen reconstruction to improve segmentation accuracy of tissue compartments (stroma, lumen, epithelium), achieving average Dice score of 0.895.
- Implementing and evaluating deep learning and vision foundation architectures (UNet, nnUNet, UNI) for robust multi-class segmentation across diverse histopathological datasets.
- Extending research to Gleason pattern grading (patterns 3, 4, 5) to enable deep learning automated cancer aggressiveness assessment.

Research Project, NLP and LLMs (University of Southern California) Aug. - Dec. 2025
Paper

- Used multiple unlearning algorithms (DPO, refusal tuning, and prompting) to remove political information from different LLMs (Llama-3.1-8B & Mistral-7B) to see the effects.
- These effects were measured using political compass and social psychology questionnaires (Pew Research Center Political Typology Test, Right-Wing & Left-Wing Authoritarianism, Social Dominance Orientation, and Openness to Experience in the Big Five).
- Specifically, if we remove information of politicians of a particular political party, does this shift the LLMs observed "political orientation" away from that ideology?

Honors Thesis, Dr. Daniel Myers (Rollins College) Jan. – Dec. 2023
Project Paper

- Evaluated LLM performance on sentiment analysis using manually labeled real-world interview data from non-profit Crave, comparing human-coded emotional assessments to model-generated outputs.
- Investigated limitations of LLMs in capturing nuanced emotional context in low-resource, qualitative settings, providing insights for responsible deployment in social science research.

NSF REU Intern, Pennsylvania State University – **ML in Cybersecurity** June - Aug. 2023
Paper

- Investigated adversarial robustness of ML models for autonomous driving systems, analyzing vulnerabilities under perturbation attacks and evaluating transfer learning across architectures.
- Conducted interdisciplinary research, covering qualitative and quantitative methods, research ethics, and perspectives from computer science and psychology.

Technical Personal Projects

ASLearning, ML for Accessibility Jan. – May 2023
Project

- Built mobile application using real-time object detection (TensorFlow) to facilitate communication between hearing-impaired and English-speaking users through sign language recognition.

voteSmarter, Civic Engagement Mobile App Feb. – June 2024
Project

- Developed mobile application connecting users to political candidates based on values alignment and location; integrated features for voter registration, polling locations, and candidate databases.

Dog Rescue Website — Oct. 2025 – Jan. 2026
Website

- Developed a website for a dog rescue that runs completely on donations. Their only online presence previously was their Facebook page, and their growing rescue needed a bigger space to spread information about their adoptable dogs.

Publications & Submissions

Shah, A. (presenting author), Bogale, Y., Nair, V., Collins, K., Feldman, M., Bahler, C., Shiradkar, R. *"Automated gleason pattern segmentation in prostate whole slide digitized pathology using foundation model feature embeddings"*, **Annual Meeting of the American Urological Association (AUA), 2026** (interactive poster). — May 2026 **Accepted for Poster Presentation**

Shah, A. (presenting author), Bogale, Y., Murugan, N., Nair, V., Collins, K., Feldman, M., Bahler, C., Shiradkar, R., *"Automated segmentation of gland and stromal morphology on digitized H&E-stained prostate whole slide images"*, **Annual Meeting of the American Urological Association (AUA), 2026** (interactive poster). — May 2026 **Accepted for Poster Presentation**

Shah, A. (presenting author), Bogale, Y., Murugan, N., Nair, V., Collins, K., Bahler, C., Feldman, M., Shiradkar, R. *"Deep-learning-based segmentation of glands and stroma on digitized H&E-stained prostate whole slide images"*, **SPIE Medical Imaging 2026** (oral presentation). — Feb. 2026 **Accepted for Oral Presentation**

Shah, A., Hu, Y., Le, T. (2026). *"What the "Spotless" Mind Remembers: How Knowledge Entanglement Shapes What Leaks After Unlearning in LLMs"* arXiv preprint arXiv:2510.25732 — Aug. 2026 **Submitted to EACL 2026**

Shah, A. (2023). *"Evaluating AI Sentiment Analysis."* Rollins College Honors Program Theses, No. 218. Available at: scholarship.rollins.edu/honors/218/ — Dec. 2023 **Honors Thesis**

Teaching and Outreach

Volunteer, Amy's Angels Dog Rescue — Sept. 2025 - Feb. 2026

- Volunteered to create a website for this dog rescue whose only online presence was Facebook.
- Needed a way to communicate their adoptable dogs and donation needs to their community.

Volunteer, Viterbi Impact — Aug. 2024 – May 2026

- Engaging in community service projects including fire clean-up, premature baby care, and food distribution.
- Helped distribute food for people who were affected by the 2025 Southern California Wildfires and the nationwide EDT crisis.

Volunteer, AAPI Coming Together (ACT) — May 2024 – Ongoing

- Supporting civic outreach and cultural advocacy within the Asian American community.
- Developed **voteSmarter** to support voter education efforts and civic engagement.

Instructor, iDTech Camps — June – Aug 2022

- Taught Java, Python, C#, and game development to K-12 students, promoting computational thinking and creative problem-solving in early learners.

Honors & Awards

Dean's List, University of Southern California (multiple semesters) — 2025-2026
REU & IRES Awardee, National Science Foundation — 2023 & 2026
Archibald Granville Bush Award: Excellent Standard in Academic Pursuit, Rollins College — 2023